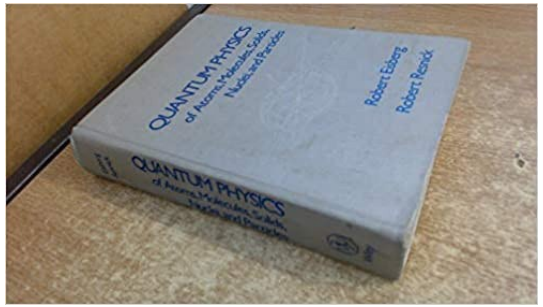
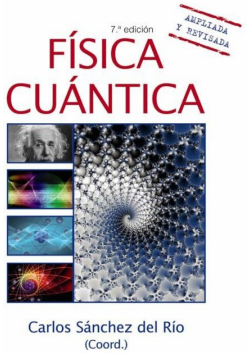
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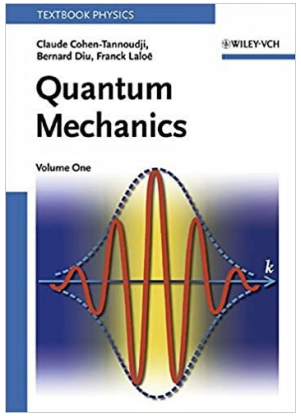
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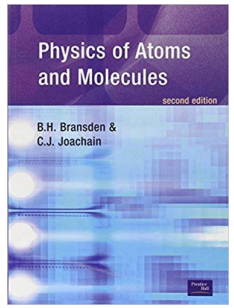
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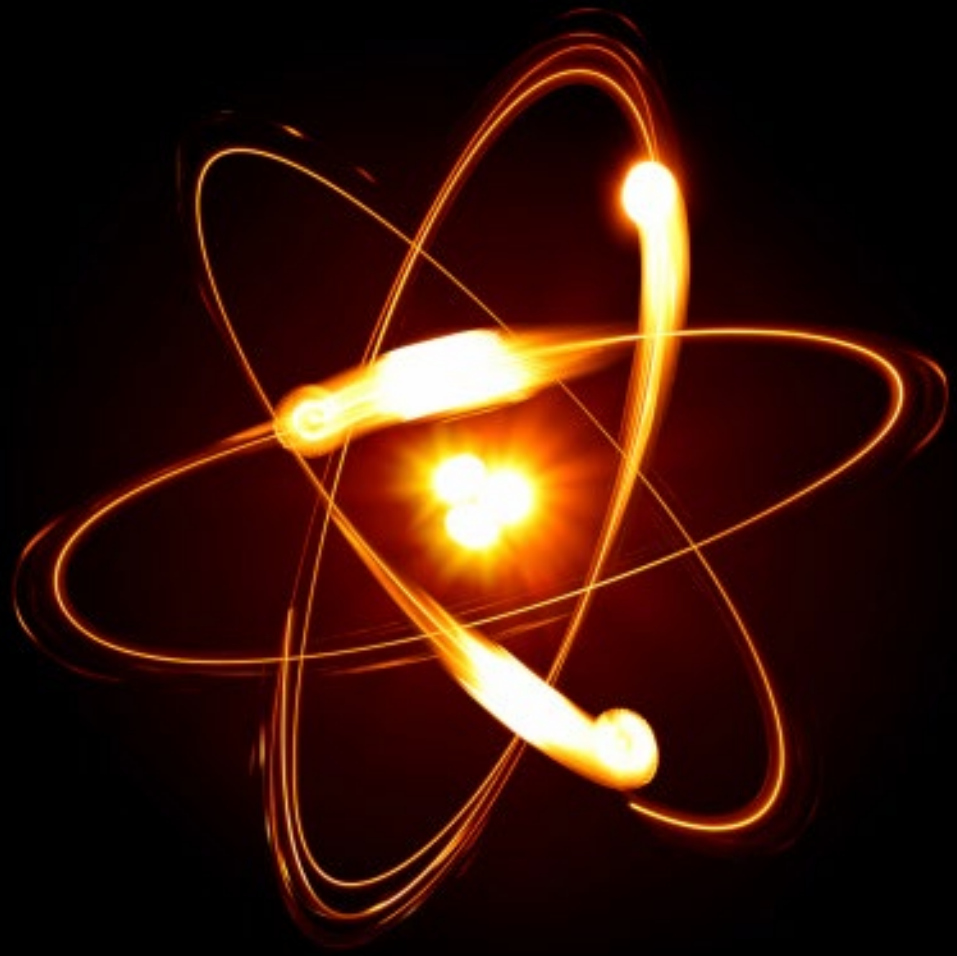
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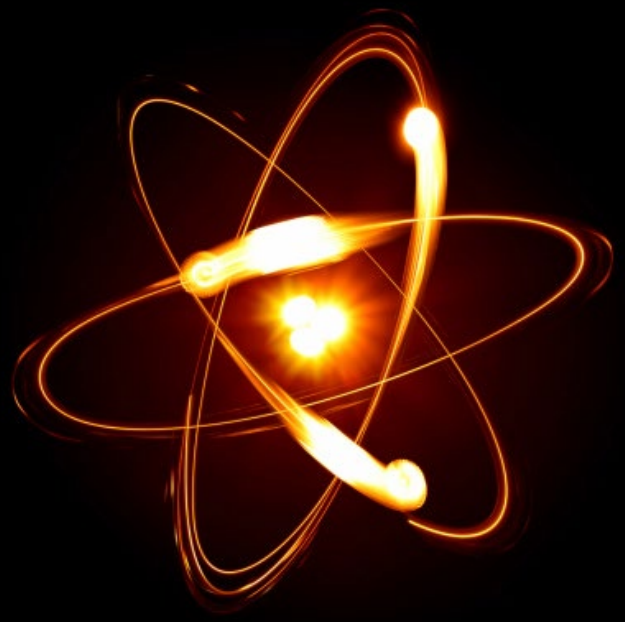
Chapter 1.

Thermal radiation and Planck´s postulate



Beatriz González del Río

Quantum Physics



Chapter 1.

Thermal radiation and Planck´s postulate

- Introduction
- Thermal radiation
- Blackbody radiation
- Classical theory of cavity radiation
- Planck’s theory of cavity radiation
- Plank’s formula
- Case study: pendulum
- Does the sun behave like a blackbody?

Introduction: particle/wave models

By the end of the XIX century, all (known) physical phenomena could be explained by resorting to two different models:

- Particle-like model → Newton's laws

  Motion of the planets, atoms, molecules…..

- Wave-like model → Maxwell's equations

  Phenomena with light (*interference, diffraction*)

Physics seemed almost complete… So what could possibly go wrong?

Some problems

…… there appeared some new "problems" whose explanation could not be achieved within the classical theory:

- **Blackbody radiation**
- Photoelectric effect
- Specific heat of solids
- Nuclear radioactive decay
- Atomic spectra
- Compton effect .....

Its explanation could NOT be achieved using the classical ideas.

New ideas ⇒ QUANTUM PHYSICS

Birth of Quantum Theory

The birth of the Quantum Theory can be dated on December 14, 1900 when Max Planck presented his paper "Zur theorie des Gesetzes der Energieverteilung im Normalspektrum" " On the theory of the Energy Distribution Law of the Normal Spectrum"

Quantum Physics is a generalization of classical physics extending its range of application to the world of very small dimensions

Includes the classical laws when dealing with macroscopic magnitudes.

There will appear a new characteristic constant *h* (Planck's constant)

$$h = 6.626 \cdot 10^{-34} Joules \cdot second$$



Thermal radiation

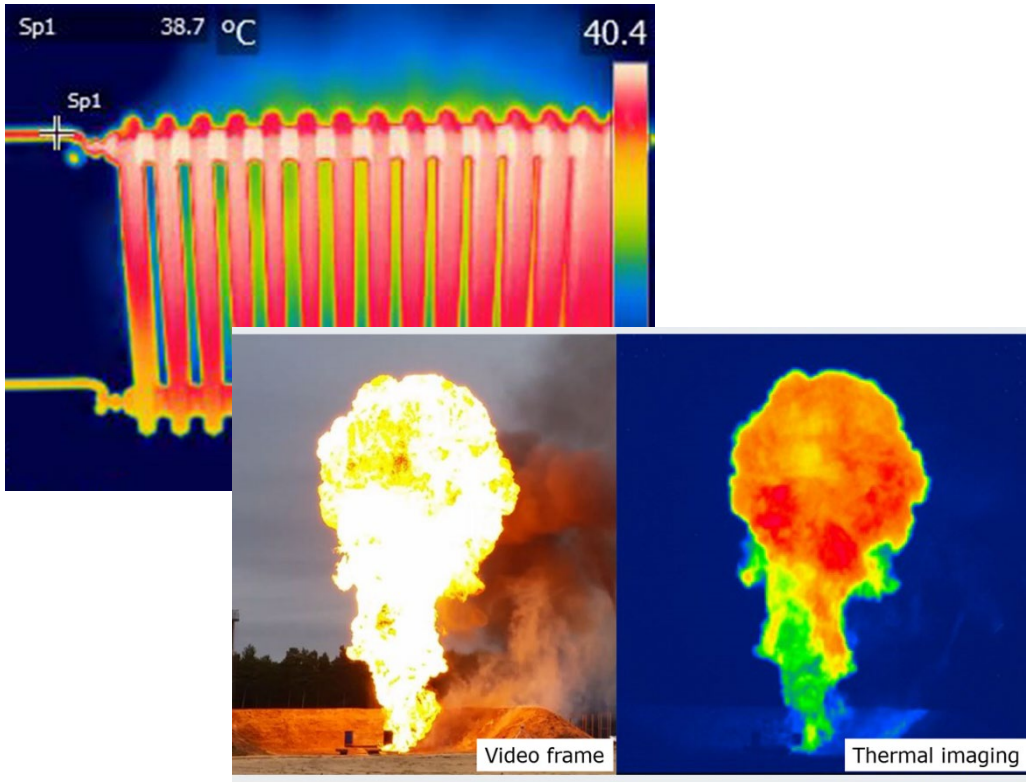
Thermal radiation is the electromagnetic radiation emitted by a body as a result of its temperature $(T \neq 0)$.

All bodies emit and absorb

Thermal equilibrium → **Both rates are equal**

Frequencies outside the visible range (for example, at room temperature).

Strongly dependent on the temperature.



Thermal radiation

At ambient temperature (room temperature) most bodies are visible due to the light they reflect (NOT because of their emitted light).

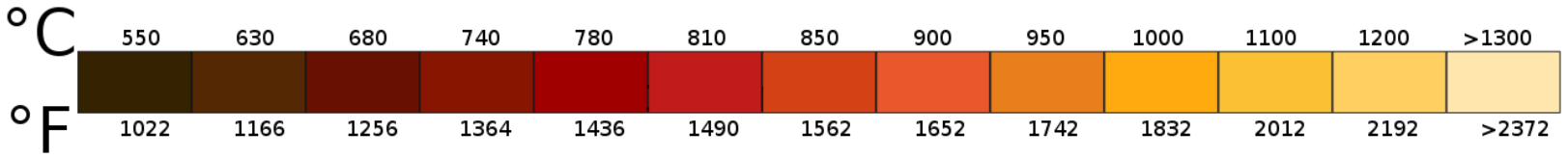
No reflected light? We see NOTHING

At very high temperatures (several thousand degrees) bodies are self-luminous (but still around 90% is infrared).

Example: Heating an iron poker in a fire

Low temperature → Radiates heat (you can feel it) BUT NOT visible light

Increase temperature → Visible effects:





Thermal radiation

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Increase temperature → Visible effects: dull red colour, bright red colour....., intense blue-white colour (at very high T)

With increasing T the body emits more thermal radiation

Thermal radiation

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Example: Heating an iron poker in a fire

Low temperature → Radiates heat (you can feel it) BUT NOT visible light

Increase temperature → Visible effects: dull red colour, bright red colour....., intense blue-white colour (at very high T)

With increasing T the body emits more thermal radiation and the frequency of the most intense radiation becomes higher.

There is a continuous spectrum

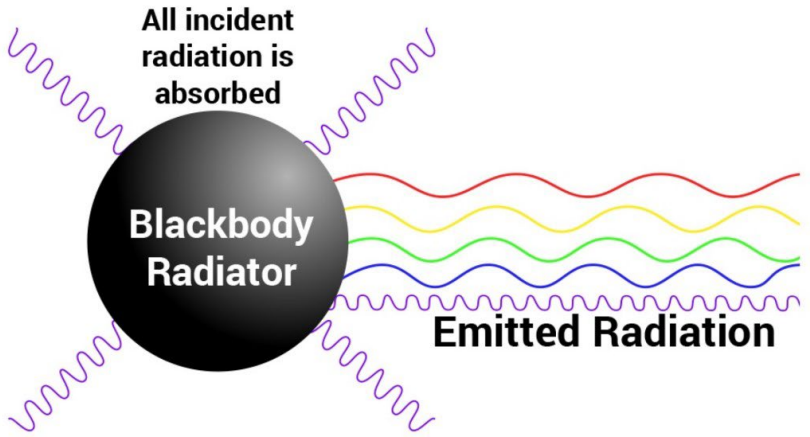
Blackbody radiation

Generally, when radiation impinges on a body, partly is absorbed and partly is reflected.

Special class of bodies: Blackbody

A blackbody absorbs all the radiation incident upon it.

Example: Any object coated with a layer of black pigment



Moreover, independently of their composition, all blackbodies at the same temperature emit thermal radiation with the same spectrum.

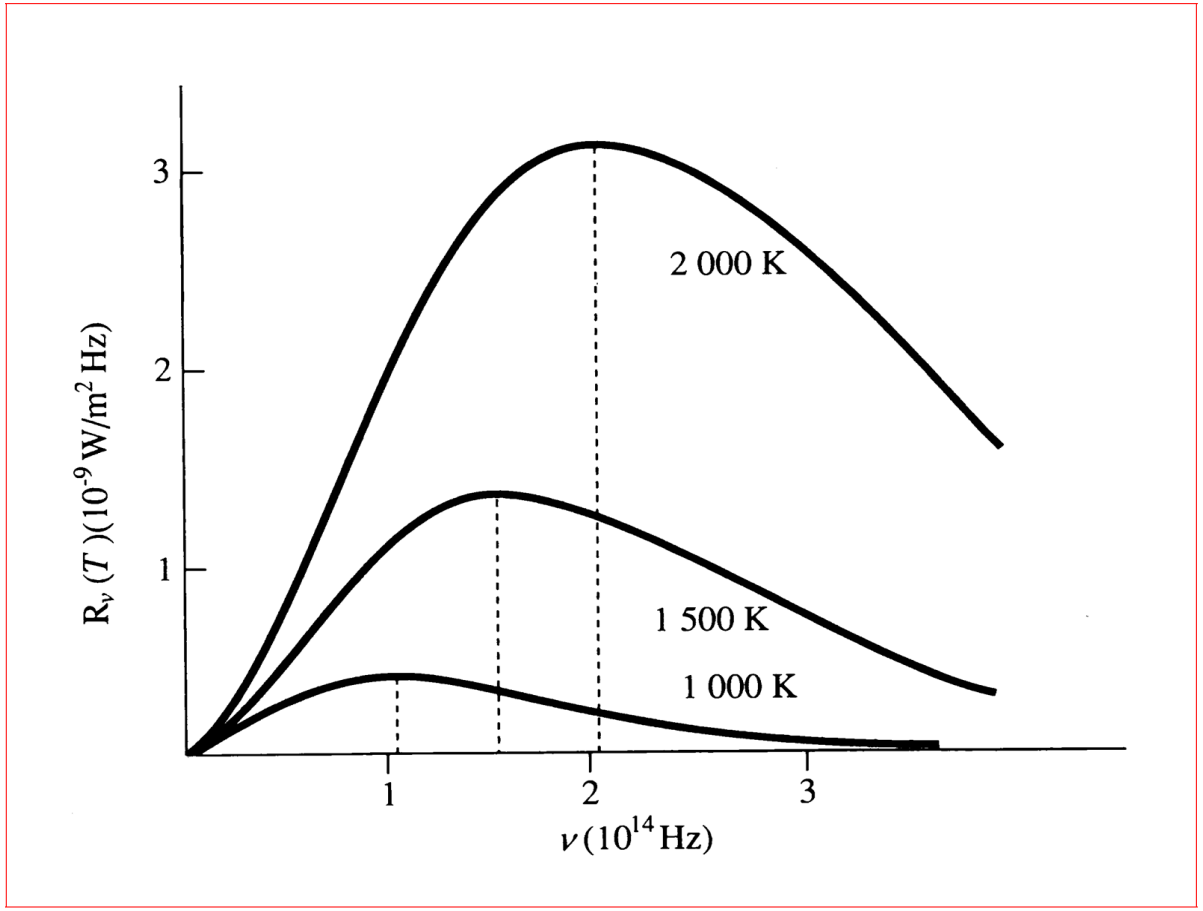
This **universal character** of the radiation emitted by blackbodies makes them of particular theoretical interest.

How to quantify the blackbody radiation?

Spectral radiancy, $R_T(\nu)$:

$R_T(\nu)\,d\nu$: energy emitted per unit time in radiation of frequency in the interval ν, $\nu + d\nu$ from a unit area of a surface at temperature T.

First accurate measurements of $R_T(\nu)$ were performed by Lummer and Pringsheim (1899)



How to quantify the blackbody radiation?

- Radiancy, → $R_T = \int_0^\infty R_T(\nu)\, d\nu$

  total energy emitted per unit time and unit area from a blackbody at temperature T.

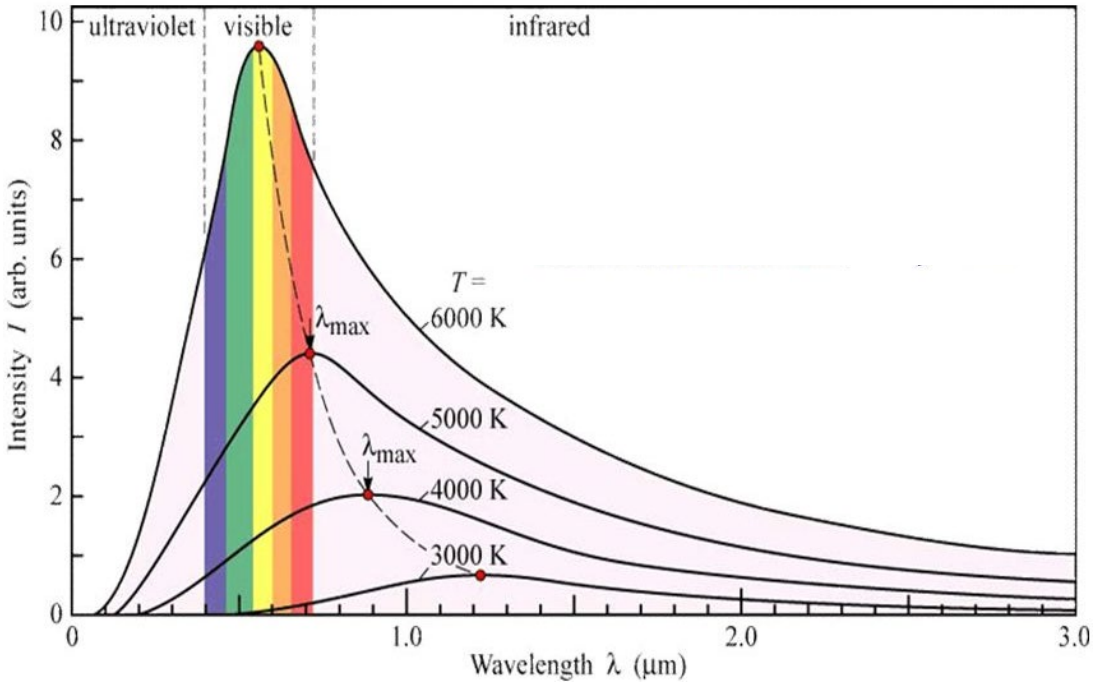
- Stefan's law (1879) $R_T = \sigma T^4$

  Stefan-Boltzmann constant $\sigma = 5.67 \cdot 10^{-8}\, W/m^2K^4$

- Wien's displacement law (1893) $\nu_{max} \propto T$

  $\lambda_{max}\, T = constant \longrightarrow constant = 2.8978 \cdot 10^{-3} mK$

If T doubles, what happens to λ_{max} and total emitted power?



How to quantify the blackbody radiation?

- Radiancy, → $R_T = \int_0^\infty R_T(\nu)\, d\nu$

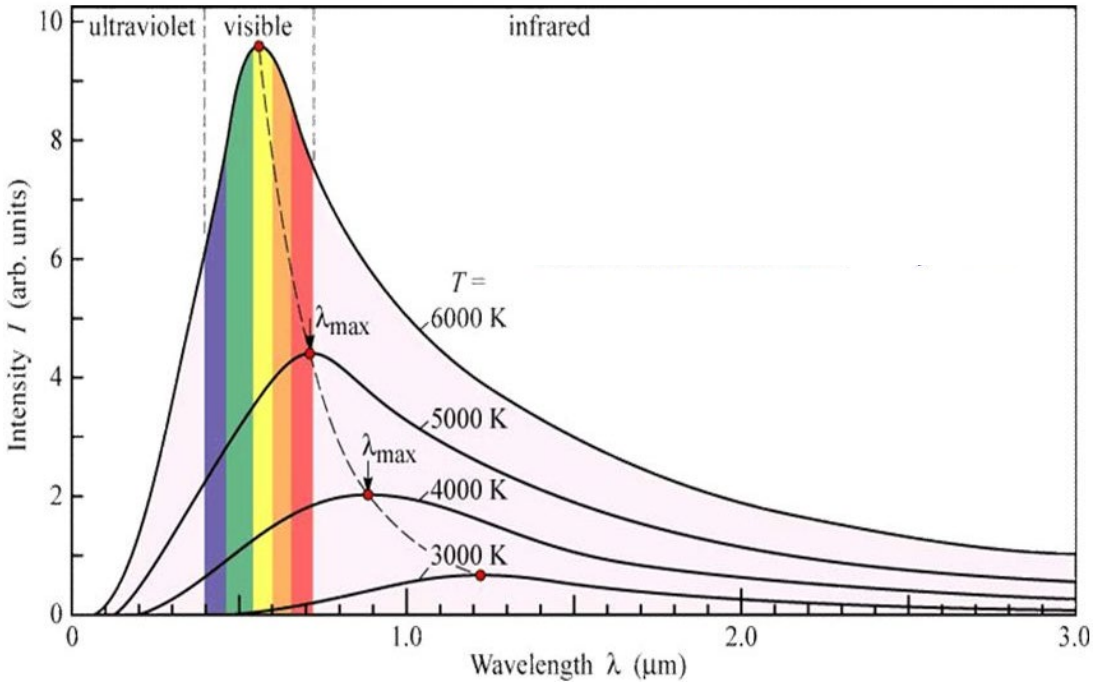
  total energy emitted per unit time and unit area from a blackbody at temperature T.

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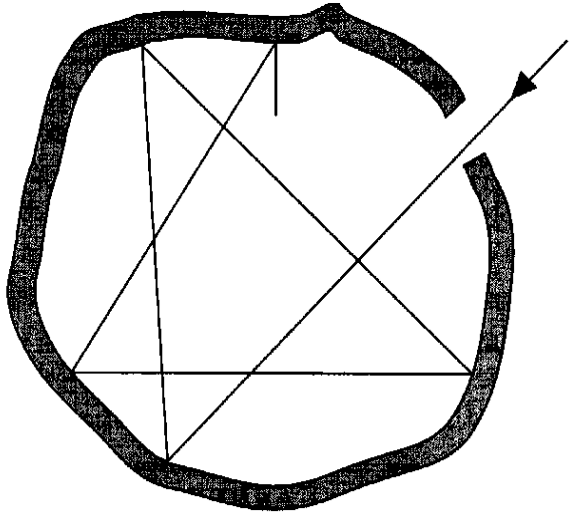


Assuming that stellar surfaces behave like blackbodies :

The Sun	λ_{max} = 5100 Å	T= 5700 K
The North Star	λ_{max} = 3500 Å	T = 8300 K

Cavity blackbody

Another example of blackbody: Object with a cavity which is connected to the outside by a small hole



Why can a tiny hole behave almost like a perfect absorber?

Radiation incident upon the hole (very small) enters the cavity and is reflected back and forth until it is completely absorbed by the walls. The hole behaves like a blackbody.

Now, the walls are heated to a temperature T. They emit thermal radiation which fills the cavity (we have something like a sea of radiation).

Radiation leaving the hole (blackbody) must have a blackbody spectrum. This radiation is a sample (same composition) of the radiation inside the cavity → The radiation inside the cavity has the spectrum of a blackbody (defined by T).

Blackbody radiation

Spectrum inside the cavity is characterized by $\rho_T(\nu)$ (spectral energy density).

$\rho_T(\nu)d\nu$: Energy contained in a unit volume of the cavity at a temperature T and in the frequency interval ν to $\nu + d\nu$

It can be shown that

$$R_T(\nu) = \frac{c}{4}\rho_T(\nu)$$

Performing the integral in ν

$$R_T = \int_0^\infty R_T(\nu)d\nu = \frac{c}{4}\int_0^\infty \rho_T(\nu)d\nu = \frac{c}{4}\rho_T$$

where $\rho_T(\nu)$ stands as the magnitude to be determined in order to explain the blackbody spectrum.

This was one of the main theoretical problems at the end of the 19th century.

Classical theory of cavity radiation

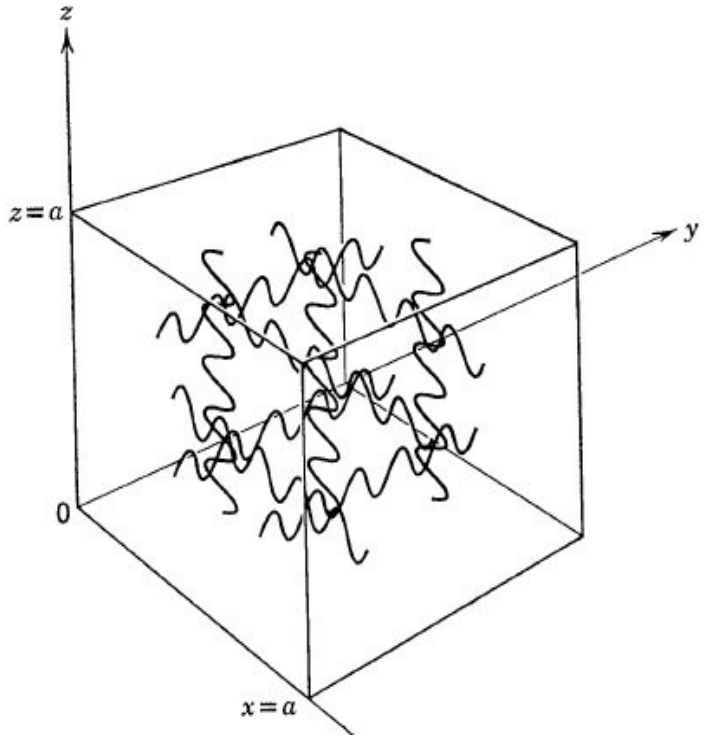
Can classical physics predict the blackbody spectrum?

Rayleigh and Jeans (1900) made a calculation of the spectral energy density of cavity radiation.

Their results did NOT agree with experimental data.

Calculation was based on classical ideas.

Cavity with walls at temperature T → emit electromagnetic radiation.



Classical theory of cavity radiation

Consider a cavity whose walls are at temperature T → emit electromagnetic radiation.

1) Mode counting (Classical electromagnetic theory): The radiation inside the cavity exists in the form of electromagnetic standing waves with nodes at the walls. The number of standing waves within the frequency interval *ν* to *ν* + *dν*

$$N(\nu)d\nu = (8\pi V\nu^2/c^3)d\nu$$

2) Equipartition (Classical kinetic theory): The average total energy per standing wave (law of equipartition of energy)

$$\bar{\epsilon} = k_B T$$

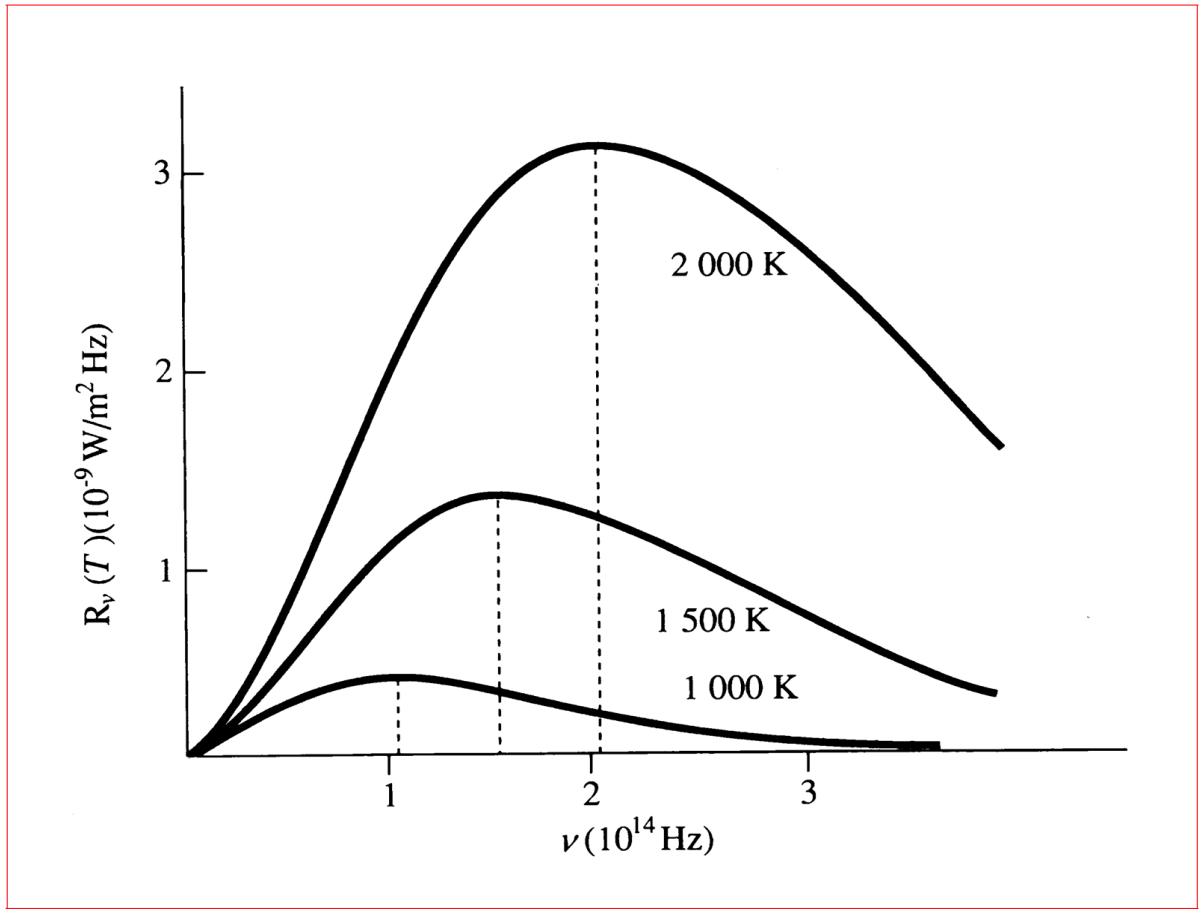
3) Combine

$$\rho_T(\nu) = \frac{N(\nu)\bar{\epsilon}}{V} = \frac{8\pi\nu^2 k_B T}{c^3}$$

Classical theory of cavity radiation

$$\rho_T(\nu)d\nu = \frac{N(\nu)\bar{\epsilon}}{V}d\nu = \frac{8\pi\nu^2 k_B T}{c^3}d\nu$$

DOES IT WORK???



Classical theory of cavity radiation

$$\rho_T(\nu)d\nu = \frac{N(\nu)\bar{\epsilon}}{V}d\nu = \frac{8\pi\nu^2 k_B T}{c^3}d\nu$$

Diverges as ν^2.
Ultraviolet catastrophe

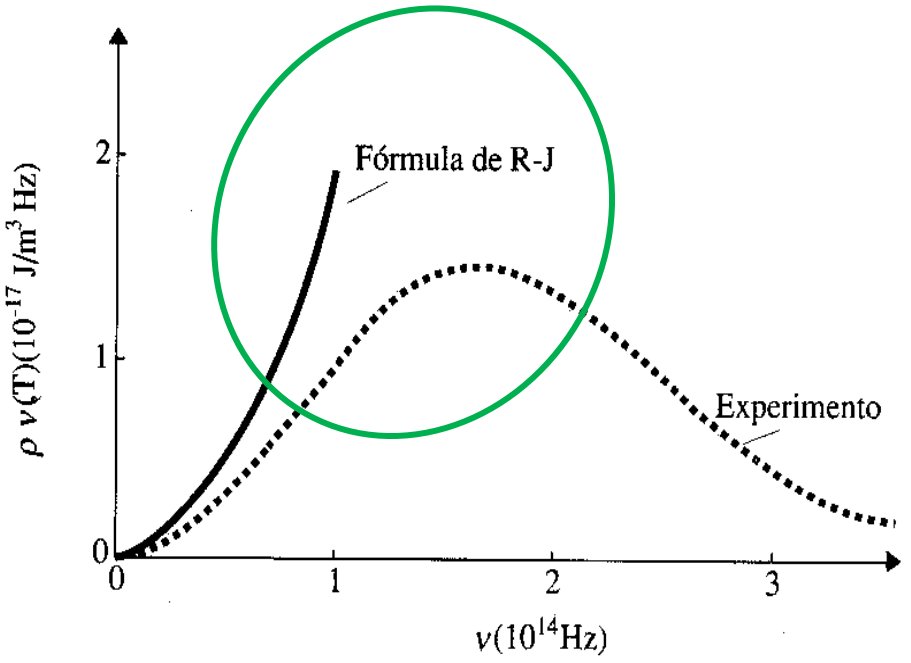


Figura 1.8. Comparación de las previsiones de la fórmula de Rayleigh-Jeans con los resultados experimentales: catástrofe ultravioleta.

Planck's theory of cavity radiation

We have $\rho_T(\nu)d\nu = N(\nu)\bar{\epsilon}\frac{1}{V}d\nu$ (it works when $\nu \to 0$)

Something in the classical derivation must fail. Which ingredient would you question?

A) EM standing waves B) mode counting C) $\bar{\epsilon} = k_B T$

Planck assumed that the problem arose from the expression for $\bar{\epsilon}$.

$$\begin{cases} \bar{\epsilon} \longrightarrow k_B T & \text{when } \nu \longrightarrow 0 \\ \bar{\epsilon} \longrightarrow 0 & \text{when } \nu \longrightarrow \infty \end{cases}$$

Planck suggested that a dependence on the frequency should be introduced, namely $\bar{\epsilon} \longrightarrow \bar{\epsilon}(\nu)$

Classical assumption: energy is continuous

Let's consider a system composed of a large number of identical entities.

The system is isolated so its total energy is a constant but the entities can exchange energy among them.

According to the classical statistical mechanics the energies of the entities are distributed according to the *Boltzmann distribution function*,

$$P(\epsilon) \propto e^{-\epsilon/k_B T}$$

$P(\epsilon)d\epsilon$ proportional to the probability of finding an entity with energy within the interval ϵ and $\epsilon + d\epsilon$

Average energy (expected value)

$$\bar{\epsilon} = \frac{\int_0^\infty \epsilon P(\epsilon) d\epsilon}{\int_0^\infty P(\epsilon) d\epsilon} = \frac{-(k_B T + \epsilon) e^{-\epsilon/k_B T}\Big]_0^\infty}{-e^{-\epsilon/k_B T}\Big]_0^\infty} = k_B T$$

Notice that ϵ plays the role of a continuous random variable.

Planck's theory of cavity radiation

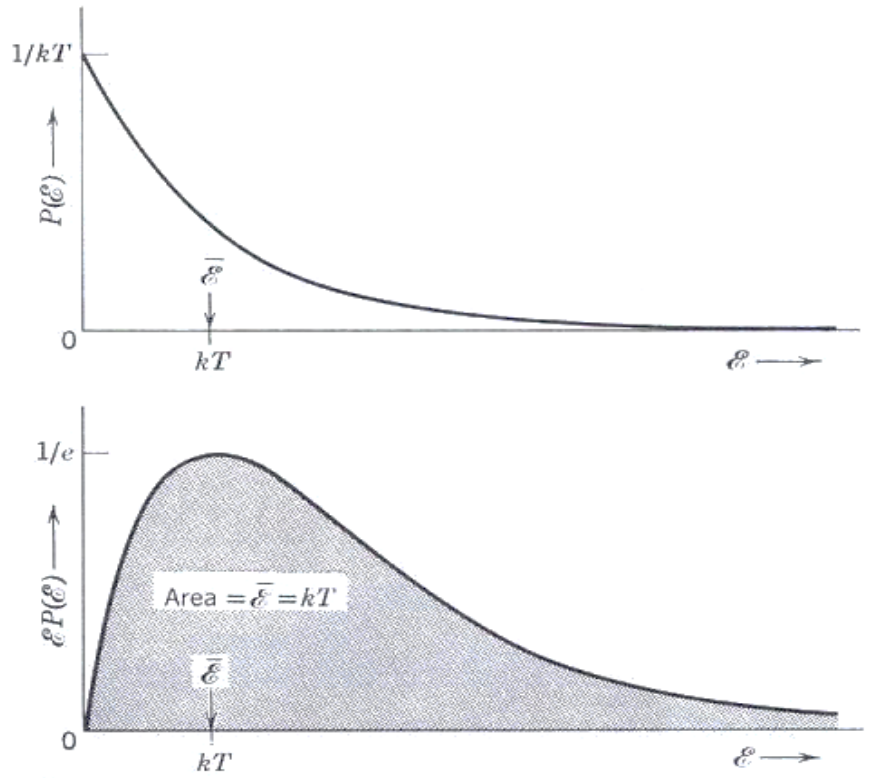


FIGURE I–9

Top: A plot of the Boltzmann probability distribution $P(\mathscr{E}) = e^{-\mathscr{E}/kT}/kT$. The average value of the energy $\mathscr{E}$ for this distribution is $\bar{\mathscr{E}} = kT$, which is the classical law of equipartition of energy. To calculate this value of $\bar{\mathscr{E}}$, we integrate $\mathscr{E}P(\mathscr{E})$ from zero to infinity. This is just the quantity that is being averaged, $\mathscr{E}$, multiplied by the relative probability $P(\mathscr{E})$ that the value of $\mathscr{E}$ will be found in a measurement of the energy. *Bottom:* A plot of $\mathscr{E}P(\mathscr{E})$. The area under this curve gives the value of $\bar{\mathscr{E}}$.

Planck's theory: discrete energy packets

Planck realized that he could reproduce the experimental data IF the variable ϵ was considered a **discrete** (instead of continuous) random variable.

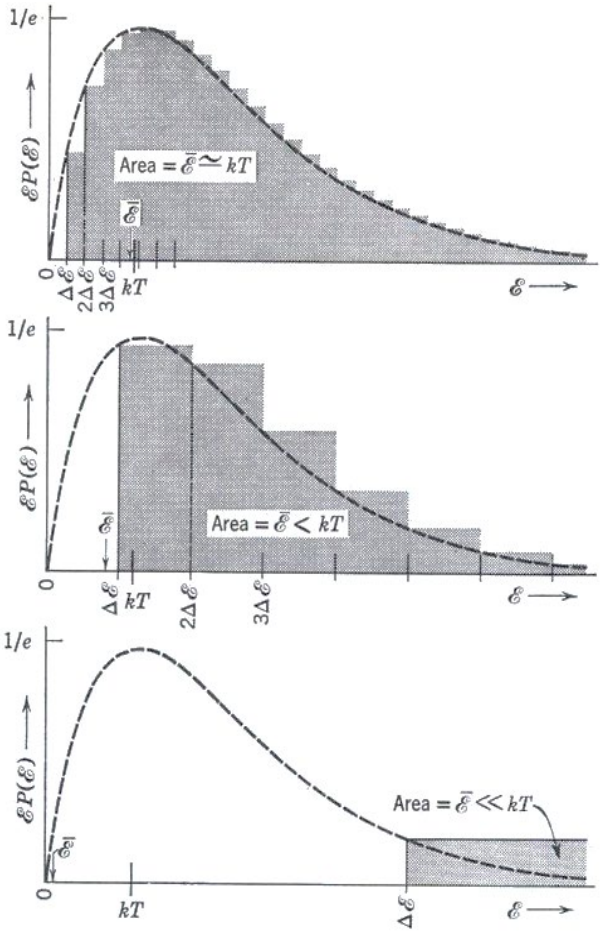
Assumes that $\epsilon = \epsilon_0, \epsilon_1, \epsilon_2, \ldots, \epsilon_n, \ldots$ $\quad\quad$ $\epsilon_n = n\Delta\epsilon$

Planck uses the same expression for $P(\epsilon)$

The mean value: $\bar{\epsilon} = \dfrac{\sum_{n=0}^{\infty} \epsilon_n P(\epsilon_n)}{\sum_{n=0}^{\infty} P(\epsilon_n)} = \dfrac{\sum_{n=0}^{\infty} \epsilon_n P(\epsilon_n)\Delta\epsilon}{\sum_{n=0}^{\infty} P(\epsilon_n)\,\Delta\epsilon}$

Three cases: $\Delta\epsilon \ll k_B T$, $\quad$ $\Delta\epsilon \approx k_B T$, $\quad$ $\Delta\epsilon \gg k_B T$

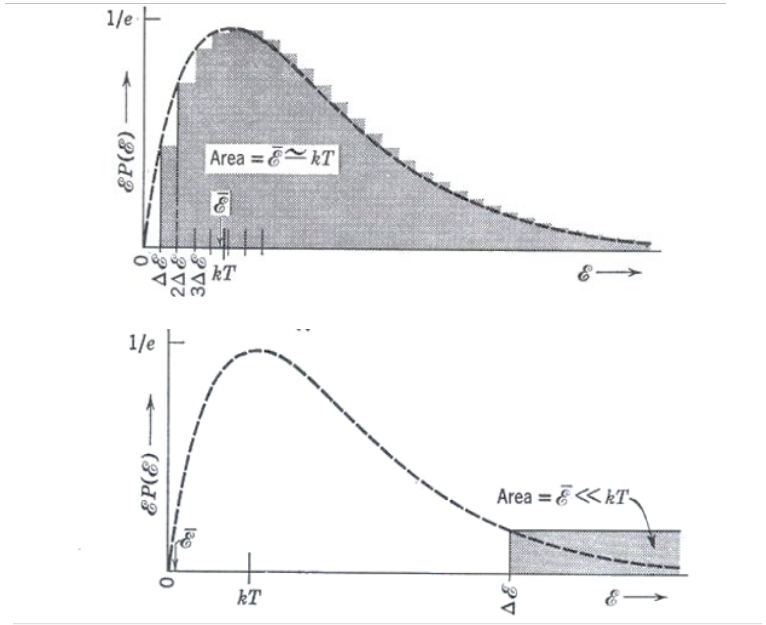
$$\bar{\epsilon} = \frac{0 + \epsilon_1 P(\epsilon_1)\Delta\epsilon + \epsilon_2 P(\epsilon_2)\Delta\epsilon + \cdots}{P(\epsilon_0)\Delta\epsilon + P(\epsilon_1)\Delta\epsilon + \cdots}$$



Planck's theory of cavity radiation

$\Delta\epsilon \ll k_B T$, then $\bar{\epsilon} \longrightarrow k_B T$ (valid when $\nu \longrightarrow 0$)

$\Delta\epsilon \gg k_B T$, then $\bar{\epsilon} \longrightarrow 0$ (valid when $\nu \longrightarrow \infty$)



Planck's leap: $\Delta\epsilon = h\nu$ with h an unknown proportionality constant whose value is found by fitting to experiment.

$$h = 6.626\ 10^{-34}\ J \cdot s$$

Planck did NOT modify Boltzmann's distribution function!

Planck assumed that the distribution function was for a discrete random variable

Planck's theory of cavity radiation

Calculating $\bar{\epsilon}$: According to Planck the energy of a standing wave, can only take discrete values, so, $\epsilon = \epsilon_0, \epsilon_1, \epsilon_2, \dots, \epsilon_n, \dots$

Planck proposed $\Delta\epsilon = h\nu$ $\epsilon_n = n\Delta\epsilon = \mathrm{n}h\nu$

Then

$$\bar{\epsilon}(\nu) = \frac{\sum_0^\infty \epsilon_n P(\epsilon_n)}{\sum_0^\infty P(\epsilon_n)} = \cdots = \frac{h\nu}{e^{h\nu/k_B T} - 1}$$

Notice that it recovers the correct limits $\nu \longrightarrow 0$ and $\nu \longrightarrow \infty$

Planck's theory of cavity radiation

Planck's blackbody spectrum

$$\rho_T(\nu) = \frac{8\pi\nu^2}{c^3}\frac{h\nu}{e^{h\nu/k_BT} - 1}$$

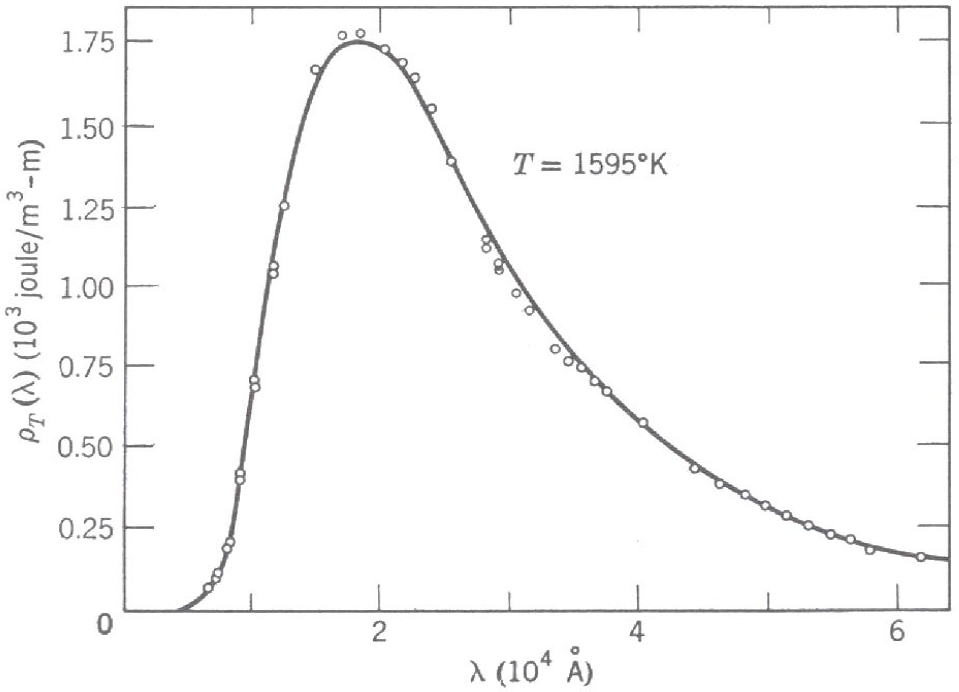


FIGURE I–11

In terms of the wavelength λ,

$$\rho_T(\lambda)d\lambda = -\rho_T(\nu)d\nu$$

$$\left(\nu = {}^{c}/_{\lambda}\right)$$

$$\rho_T(\lambda) = \frac{8\pi hc}{\lambda^5}\frac{1}{e^{hc/\lambda k_BT} - 1}$$

Planck's theory of cavity radiation

Stefan's law (1879)

$$R_T = \sigma T^4 \quad \longrightarrow \sigma = 5.67 \cdot 10^{-8}\ {}^{W}/_{m^2K^4}$$

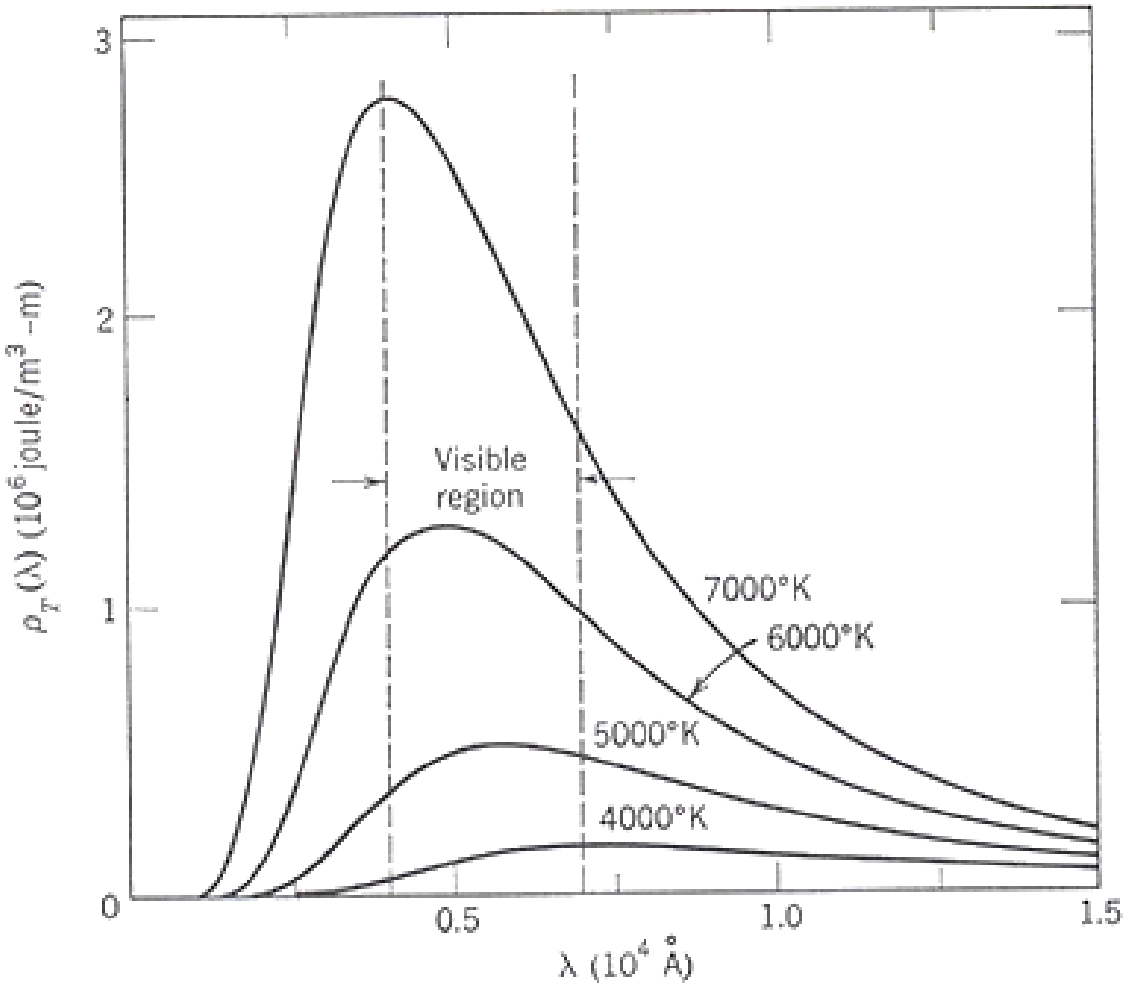
Now

$$R_T = \int_0^\infty R_T(\nu)d\nu = \int_0^\infty \frac{c}{4}\rho_T(\nu)d\nu = \frac{2\pi^5 k_B^4}{15c^2h^3}T^4$$

Wien's law

$$\left.\frac{d\rho_T(\lambda)}{d\lambda}\right|_{\lambda_{max}} = 0 \qquad \lambda_{max}T = 0.2014\ \frac{hc}{k_B}$$

Experimental data for $\rho(\lambda, T)$



The maximum moves towards smaller λ's as the T is increased.

Planck's formula

- Recovers Rayleigh-Jeans' at low frequencies

- Integrating $\rho_T(\nu)$ over all ν's gives Stefan's law and an expression for σ.

$$R_T = \sigma T^4 \quad \longrightarrow \sigma = 5.67 \cdot 10^{-8}\ {}^{W}/_{m^2K^4}$$

- Evaluating the maxima in $\rho_T(\nu)$ we obtain Wien's law.

$$\left.\frac{d\rho_T(\lambda)}{d\lambda}\right|_{\lambda_{max}} = 0 \qquad \lambda_{max}T = 0.2014\ \frac{hc}{k_B}$$

- When $h \longrightarrow 0$ Planck's formula reduces to the classical result.

Planck's formula

Consequences:

Plank assumed that the electrons in the walls of the cavity have an oscillatory motion and they are coupled to the electromagnetic radiation inside the cavity.

Through this coupling there is a connection between the energy of the standing electromagnetic waves and the energy of the electrons in the wall oscillating sinusoidally at the same frequency.

The oscillators in the walls of a blackbody

THEY CAN **NOT** EXCHANGE ANY AMOUNT OF ENERGY WITH THE STANDING WAVES

the interaction between radiation and matter is **quantized**.

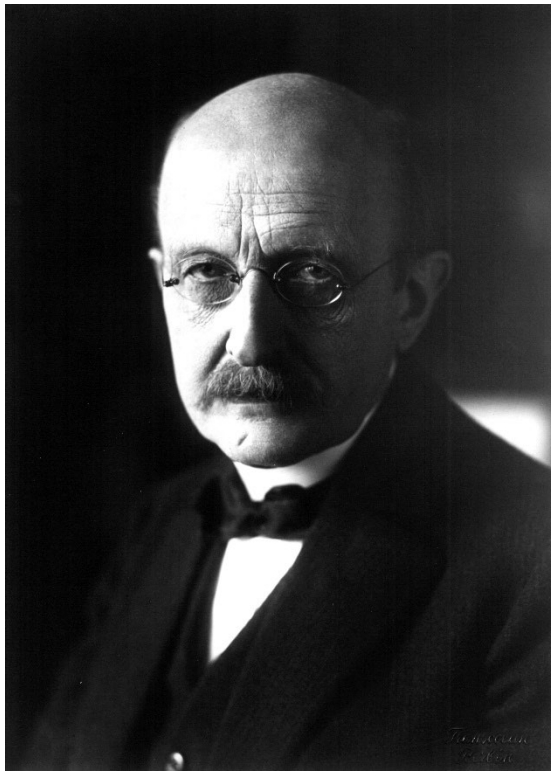
The energy is exchanged by means of bundles with energy $h\nu$, which are called quanta.

"An act of desperation"

Planck was unsure whether his introduction of the constant h was only a mathematical device or something with a deep physical significance. He wrote

I knew that the problem is of fundamental significance for physics; I knew the formula that reproduces the experimental data; so a theoretical interpretation had to be found at any cost, no matter how high.

For many years he tried to fit the quantum idea into classical theory.



Planck's formula

Starting with his study concerning the behaviour of the electrons in the walls of the cavity and their coupling to the electromagnetic standing waves inside the cavity, Planck generalized those results to any entity whose single coordinate oscillates sinusoidally.

Planck's Postulate

Any physical entity with one degree of freedom whose "coordinate" is a sinusoidal function of time (i.e. simple harmonic motion) can possess only total energies ϵ which satisfy the relation

$$\epsilon = nh\nu$$

Coordinate ⇒ Any magnitude that describes the instantaneous state of the system

Examples:

- Distance to the equilibrium position in a coil spring
- Angle of a pendulum
- Amplitude of the electric field

Planck's formula

Planck suggests a very peculiar interpretation of the oscillator's emission process (from the book "The Theory of the Heat Radiation" by Max Planck, 1914)

its emission we have, in accordance with Sec. 147, the following law: The oscillator emits in irregular intervals, subject to the laws of chance; it emits, however, only at a moment when its energy of vibration is just equal to an integral multiple n of the elementary quantum $\epsilon = h\nu$, and then it always emits its whole energy of vibration $n\epsilon$.

We may represent the whole process by the following figure in which the abscissæ represent the time t and the ordinates the energy

$$U = n\epsilon + \rho, \ (\rho < \epsilon) \qquad (251)$$

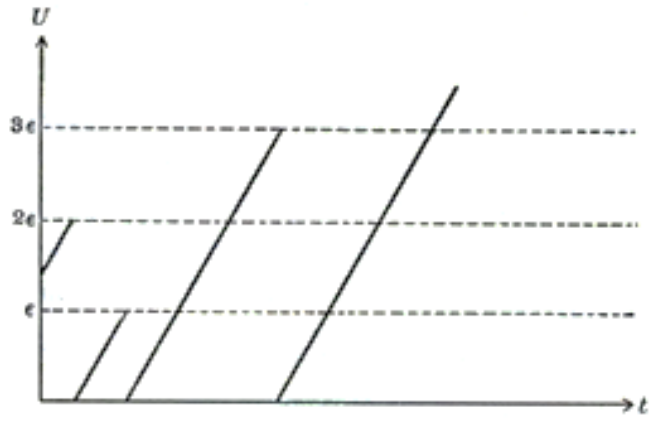
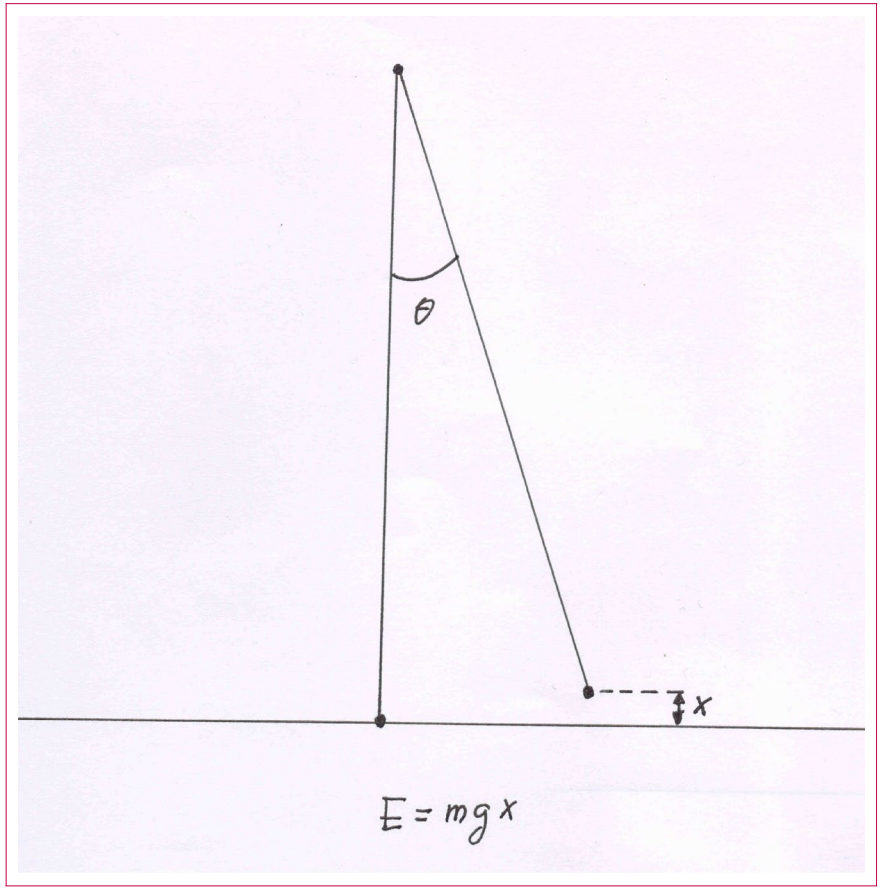


FIG. 7.

Case study: pendulum

Why has never been observed the discreteness of the energy changes in an ordinary pendulum executing simple harmonic oscillations?

Consider a point particle of 1 g which is at one end of a string whose length is 1 meter and it is oscillating in the gravitational field. Assume that the angle corresponding to the maximum separation from the vertical is one degree. The energy of the pendulum decreases due to, for instance, frictional effects. Is the observed energy decrease continuous or discontinuous ?



Note that the oscillation frequency of the pendulum is

$$\nu = \frac{1}{2\pi}\sqrt{\frac{g}{l}}$$

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$$\nu = \frac{1}{2\pi}\sqrt{\frac{g}{l}} = 0.4982\ s^{-1}$$

The energy of the pendulum is its maximum potential energy

$\mathrm{E} = \mathrm{mgx} = nh\nu$ $E = l(1 - \cos\theta) = 1.523 \cdot 10^{-4}$ m $E = 1.4925 \cdot 10^{-6}$ J

But, according to Planck' postulate, the energy is quantized, so changes occur in discontinuous jumps of magnitude $\Delta\mathrm{E} = h\nu$ with

$\Delta\mathrm{E} = h\nu = 3.30107 \cdot 10^{-34}$ J therefore $\Delta\mathrm{E}/\mathrm{E} = 2.2 \cdot 10^{-28}$

which means that we have to measure the energy with an accuracy of at least 10^{-28}

NO WAY !!!!!!!

Does the Sun behave like a blackbody?

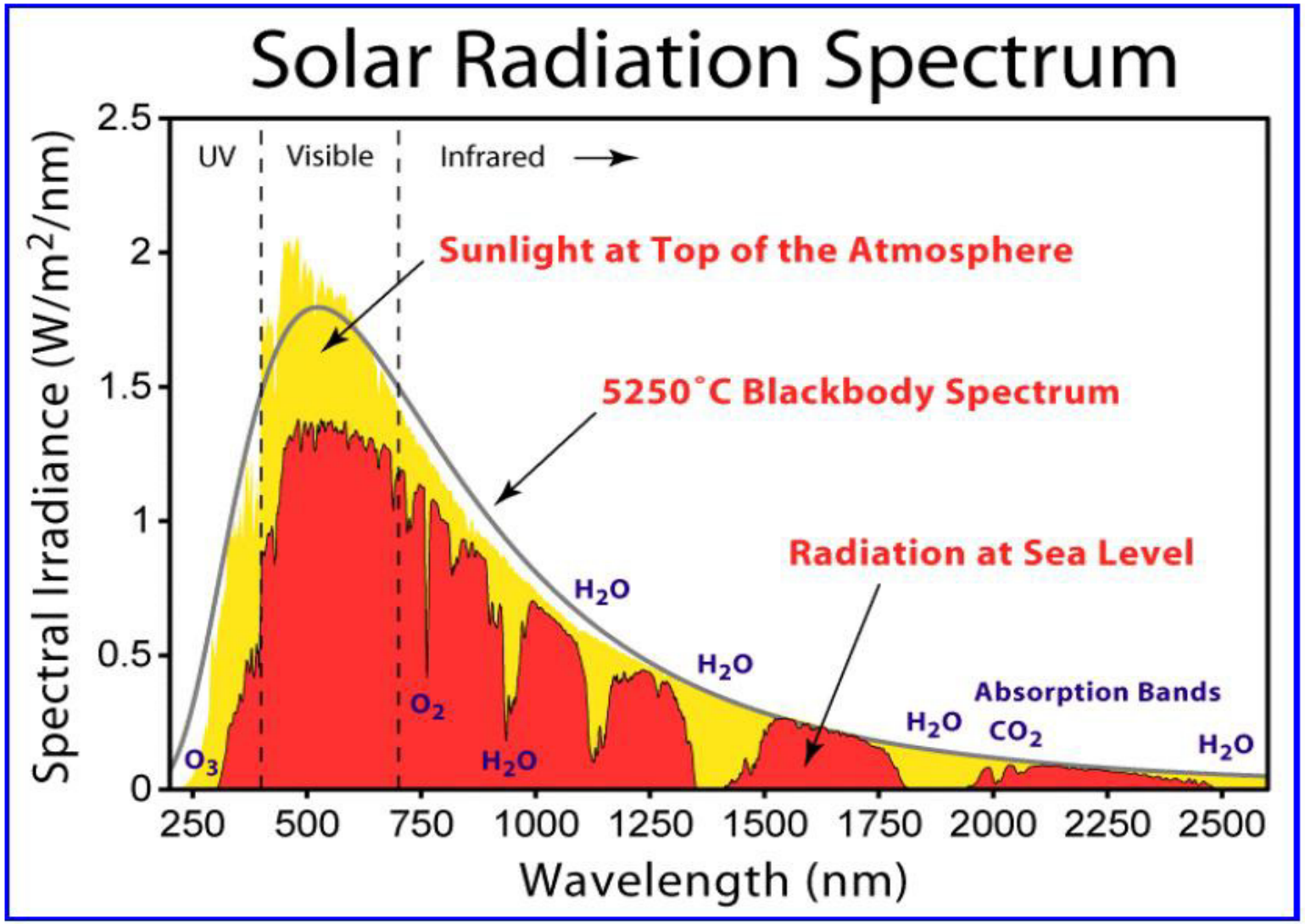


Fig.2 The solar spectrum

Stars as blackbodys

